

3) What are the types of bias in a sample survey?

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28.5 TYPES OF SAMPLING

We proceed by partitioning sampling techniques into two groups:

Non-probability (non-random) sampling

Probability (random) sampling.

A. Non-probability sampling is used in the absence of a sampling frame and are characterized by the following properties:

- 1) Some population units have no chance (zero probability) of being selected.
- 2) Units which can be selected have an unknown (non-zero) probability of selection.
- 3) Sampling errors cannot be quantified.

There are three types of non-probability sampling:

- i) **Convenience Sampling** – samples are selected from the members of the population which happen to be in the vicinity. Of course, such an approach is unlikely to yield a representative sample. For example, media is approaching people in a market for some debated political opinion. In such sampling, there is no sampling frame and sample is collected as per the convenience. People who accept the offer and give some opinion cannot be generalized because people who did not accept the offer of providing their opinion are completely ignored.
- ii) **Judgement Sampling** - uses expert opinion to judge which individuals to choose. An example would be a research firm selecting people to participate in a focus group by telephoning people and identifying who matches the target profile through a series of questions developed by an expert. Similarly in a popular TV show people from different expertise are called to discuss their opinion.
- iii) **Quota Sampling** - attempts to obtain a representative sample by specifying quota controls on certain specified characteristics, such as age, gender, social class and any other variables relevant to the investigation being undertaken. The (approximate) distribution of such characteristics in the population is required in order to replicate it in the sample. Such sampling is useful in the *absence of a sampling frame*. Since non-probability sampling technique does not require a sampling frame, it does not allow us to measure the *sampling error*.

All three techniques above (convenience, judgement and quota) have some appeal. However, in all cases we have no real guarantee that we have achieved an adequately representative sample (we might do by chance, of course, but this would be highly unlikely). Since we do not know the probability that an individual will be selected for the survey, the basic rules of inferential statistics which we will later learn to use (like confidence intervals and hypothesis testing) are not applicable here. Specifically, standard errors (a key ingredient in inferential procedures) are not estimatable. However, in the absence of a sampling frame, we will have to resort to non-probability sampling methods.

B. Probability sampling means that every population unit has a known (not necessarily equal), non-zero probability of being selected in the sample. In all cases, selection is performed through some form of randomization. For example, a pseudo-random number generator can be used (like *Tippet's random numbers tables* or by throwing of a dice, draw of a lottery, etc.) to generate a sequence of 'random' numbers. Suppose we want to select r candidates out of n , we assign the numbers 1 to n , one number to each candidate and write this numbers (1 to n) on n slips which are made as homogeneous as possible, in shape, size, etc. These slips are then put in a bag and thoroughly shuffled and the r slips are drawn one by one. The r candidates corresponding to the number on the slips drawn will constitute the random sample.

Relative to non-probability methods, probability sampling survey can be expensive and time-consuming, and requires a sampling frame. However, we can aim to minimize both the (random) sampling error and the systematic sampling bias. Since the probability of selection is known, standard errors can be computed which allows confidence intervals to be determined and hypothesis tests to be performed.

Note: *Tippet's random number tables* consist of 10400 four-digit numbers, giving in all 10400×4 , i.e., 41600 digits, taken from the British census reports. These tables have proved to be fairly random in character. Any page of the table is selected at random and the number in any row or column or diagonal selected at random may be taken to constitute the sample.

Following are the different types of probability sampling:

- i) **Simple Random Sampling (SRS)** -It is a special case where each population unit has a known, equal and non-zero probability of selection. Of the various probability samples, it is simple and it produces unbiased estimates, although more accurate methods exist (i.e., those with smaller standard errors).
- ii) **Systematic Random Sampling** - Systematic sampling is a statistical method involving the selection of elements from an ordered sampling frame. The most common form of systematic sampling is an equi-probability method. The sampling starts by selecting an element from the list at random and then

every k th element in the frame is selected, where k , is the sampling interval: this is calculated as: $k = N/n$, where n is the sample size, and N is the population size. Systematic sampling is to be applied only if the given population is logically homogeneous, because systematic sample units are uniformly distributed over the population. The researcher must ensure that the chosen sampling interval does not have a pattern. Any pattern would threaten randomness.

iii) **Stratified Random Sampling-** This sampling technique achieves a higher level of accuracy (smaller standard errors) by exploiting natural groupings within the population. Here the entire heterogeneous population is divided into a number of homogeneous groups, usually termed as '*strata*', which differ from one another but each of these groups is homogeneous within itself. Then units are sampled in random from each of these strata, the sample size in each stratum varies according to the relative importance of the stratum in the population. A simple random sample is taken from each stratum, thus ensuring a representative overall sample since a broad cross-section of the population will have to be selected. Hence great care needs to be taken when choosing the appropriate stratification factors, which should be relevant to the purpose of the sample survey such as age or gender. The sample, which is the aggregate of the sampled units of each of the stratum, is termed as stratified sample and the technique of drawing this sample is known as stratified sampling. Such a sampling is by far the best and can safely be considered as representative of the population from which it has been drawn.

iv) **Cluster Sampling** – This sampling is used to reduce costs (time and money). Here the population is divided into clusters, ideally such that each cluster is as variable as the overall population (i.e., heterogeneity within clusters and homogeneity between clusters). Next, some of the clusters are selected by SRS. This is where some savings are expected, since typically the clusters may be constructed on a geographical basis allowing interviewers to restrict themselves to visiting households in certain areas rather than having to travel long distances to meet the respondents required by a simple random sample who potentially cover a large area.

A one-stage cluster sample would mean that every unit in the chosen clusters is surveyed. It may be that the cluster sizes are large meaning that it is not feasible to survey every unit within a cluster. A two-stage cluster sample involves taking a simple random sample from the clusters selected by SRS during the first stage. When a subsample is taken from a selected cluster, this is known as a multistage design.

v) **Multistage Sampling-** It refers to the case when sample selection occurs at two or more successive stages. Multistage sampling is frequently used in large surveys. During the first stage, large compound units are sampled (primary units). During the second stage, smaller units (secondary units) are

sampled from the primary units. From here, additional sampling stages of this type may be performed, as required, until we finally sample the basic units. As we have already seen, this technique is often used in cluster sampling (as we initially sample main clusters, then clusters within clusters, etc.) We can also use multistage sampling with a mixture of techniques. A large government survey will likely incorporate elements of both stratification and clustering at different stages.

28.6 PARAMETER AND STATISTIC

In order to avoid verbal confusion with the statistical constants of the population, viz., mean (μ), variance (σ^2), etc., which are usually referred to as ‘parameters’, statistical measures computed from the sample observations alone, e.g., mean ($\bar{x}$), variance (s^2), etc., have been termed by Professor R. A. Fischer as ‘statistic’.

As discussed in the introduction, in practice, parameter values are not known and the estimates based on the sample values are generally used. Thus, statistic, which may be regarded as an estimate of parameter, obtained from the sample, is a function of the sample values only. It may be pointed out that a statistic, as it is based on sample values and as there are multiple choices of the samples that can be drawn from a population, varies from sample to sample. These differences in the values of a ‘statistic’ are called ‘*sampling fluctuations*’. The determination of the characterization of variation (in the values of the statistic obtained from different samples of a same population) that may be attributed to chance or fluctuations of sampling is regarded as one of the fundamental problems of the sampling theory.

Note: Now onwards, μ and σ^2 will refer to the population mean and variance respectively while the sample mean and variance will be denoted by $\bar{x}$ and s^2 respectively.

Check Your Progress 2

- 1) List the important types of sampling.
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- 2) Distinguish between simple random sampling and stratified random sampling.
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- 3) Differentiate between parameter and statistic.

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- 4) Distinguish between cluster sampling and stratified sampling. Are they both one stage or multistage sampling?

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28.7 SAMPLING DISTRIBUTION OF A STATISTIC

Let a random sample of size **n** be drawn from a population of size **N**. Consider the sample variable **X** to assume the values $x_1, x_2, x_3, \dots, x_n$. Consider the sample mean: $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$

If we want to use this statistic to make inferences regarding the population mean, μ , we need to know something about the probability distribution of $\bar{x}$. The distribution of a sample statistic is known as a **sampling distribution**. Two of its characteristics are of particular interest, the **mean or expected value $E(\bar{x})$** and the **variance σ^2** or standard deviation σ . What can we say about $E(\bar{x})$, the mean of the sampling distribution of $\bar{x}$?

If we draw a sample of size 'n' from a given finite population of size 'N', then the total number of ways of drawing possible samples is:

$${}^N C_n = N! / \{n! (N - n)!\} = k, \text{ (say).}$$

Suppose, we repeatedly draw sample of size **n** from the given population, and each time record the sample mean, and then take the average of those sample means. It is unlikely that any given sample will yield a value of $\bar{x}$ that precisely equals to μ , the mean of the population from which we are drawing the sample. Due to (random) sampling error some samples will give a sample mean that exceeds the population mean, and some will give an $\bar{x}$ that falls short of μ . But if our sampling procedure is unbiased, then deviations of $\bar{x}$ from μ in the upward and downward directions should be equally likely. Therefore, on an average, they should cancel out. In that case: $E(\bar{x}) = \mu = E(X)$, or the sample mean is an **unbiased estimator of the population mean**.

Consider a simple example where the population size is $N = 3$. The members are $x_1 = 1, x_2 = 3, x_3 = 5$. So, population mean is $\mu = 3$. Suppose 2 samples are to be drawn from the population of 3. So, $n = 2$ and the number of ways it can be done is ${}^N C_n = {}^3 C_2 = 3$ possible ways.

Suppose the population mean is unknown. We use simple arithmetic mean to estimate population mean.

Let the sample 1: (1,3) and the sample mean $(\bar{x}_1) = 2$;

Sample 2:(3, 5) and the sample mean $(\bar{x}_2) = 4$

Sample 3:(1,5) and the sample mean $(\bar{x}_3) = 3$

Therefore expected value of $\bar{x} = E(\bar{x}) = \frac{\bar{x}_1 + \bar{x}_2 + \bar{x}_3}{3}$

$$= (2 + 4 + 3) / 3$$

$$= 3 = \mu \text{ (Population mean)}$$

The frequency distribution of the statistic that would be obtained if the number of samples, each of the same size ('n'), were infinite is called the 'sampling distribution' of the statistic. In the case of random sampling, the nature of the sampling distribution of a statistic can be deduced theoretically, provided the nature of the population is given, from considerations of probability theory.

Like any other distribution, a sampling distribution would have its mean, standard deviations and moment of higher orders. Of particular importance is the standard deviation, which is designated as the '**standard error**' of the statistic. For example, the standard error of the sample mean is $\sigma_x = \frac{\sigma}{\sqrt{n}}$.

As an illustration, in the next section we derive for the random sampling the means (expectations) and standard errors of a sample mean and sample proportion.

We often use 0.6745 times the standard error, as the 'probable error' of the statistic. The relevance of the probable error stems from the fact that for a normally distributed variable x with mean μ and s.d σ ,

$$P [\mu - 0.6745 \sigma \leq x \leq \mu + 0.6745 \sigma] = 0.5 \text{ (approximately).}$$

28.8 STANDARD ERROR

Let us try and answer this question. *What factors do you think are going to influence the degree of dispersion of the sample mean around the population mean?*

$$\text{SE: } \sigma_x = \frac{\sigma}{\sqrt{n}}$$

The more widely dispersed are the population values around their mean, the greater the scope for sampling error (i.e., drawing by chance an unrepresentative sample whose mean differs substantially from μ). In the denominator is the square root ($\sqrt{n}$) of the sample size, n . It makes sense that if our samples are larger, this reduces the probability of getting unrepresentative sample, and hence narrows the dispersion of $\bar{x}$. Note that, it is $\sqrt{n}$, rather than n , that enters the formula. It indicates that an increase in sample size is subject to diminishing returns, in terms of increasing the precision of the estimator. For example, increasing the sample size by a factor of four will reduce the standard error of $\bar{x}$ only by a factor of two.

The standard deviation of the sampling distribution of a statistic is known as its 'standard error', abbreviated as S.E. The standard errors of some of the well-known statistics, for large samples, are given in Table 28.1, where 'n' is the sample size, σ^2 is the population variance, and P is the population proportion, and $Q = 1-P$. n_1 and n_2 represent the sizes of two independent random samples respectively drawn from the given population(s).

Table 28.1: Standard Errors of some of the well-known Statistics (for Large Samples)

S. No.	Statistic	Standard Error
1.	Sample mean: $\bar{x}$	$\sigma/\sqrt{n}$
2.	Observed sample proportion: ' p '	$\sqrt{\frac{PQ}{n}}$
3.	Sample s.d: ' s '	$\sqrt{\sigma^2/2n}$
4.	Sample variance: s^2	$\sigma^2 \sqrt{2/n}$
5.	Sample quartile	$1.36263 \sigma/\sqrt{n}$
6.	Sample median	$1.25331 \sigma/\sqrt{n}$
7.	Sample correlation coefficient	$(1-p^2)/\sqrt{n}$, p being the population correlation coefficient
8.	Sample moments μ_3	$\sigma^3 \sqrt{96/n}$
9.	Sample moments μ_4	$\sigma^4 \sqrt{96/n}$
10.	Sample coefficient of variation (v)	$\frac{v}{\sqrt{2n}} \sqrt{1 + \frac{2v^3}{10^4}} \approx \frac{v}{\sqrt{2n}}$
11.	Difference of two sample means: $(\bar{x}_1 - \bar{x}_2)$	$\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$
12.	Difference of two sample s.d.'s: $(s_1 - s_2)$	$\sqrt{\frac{\sigma_1^2}{2n_1} + \frac{\sigma_2^2}{2n_2}}$
13.	Difference of the two sample proportions: $(p_1 - p_2)$	$\sqrt{\frac{P_1 Q_1}{n_1} + \frac{P_2 Q_2}{n_2}}$

28.8.1 Utility of Standard Error

S.E plays a significant role in the large sample theory and forms the basis of the testing of hypothesis.

- The magnitude of the standard error gives an index of the precision of the estimate of the parameter. The reciprocal of the standard error is taken as the measure of reliability or precision of the statistic.

- ii) S.E. enables us to determine the probable limits within which the population parameter may lie.

Check Your Progress 3

- 1) What are the mean and standard deviation of the sampling distribution of the mean?
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- 2) What is a standard error and why is it important?
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- 3) In a random sample of 400 students of the university teaching departments, it was found that 300 students failed in the examination. In another random sample of 500 students of the affiliated colleges, the number of failures in the same examination was found to be 300. Find out the S.E of the difference between proportion of failures (i) in the university teaching departments and (ii) in the university teaching departments and affiliated colleges taken together.
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28.9 EXPECTATION AND STANDARD ERROR OF
SAMPLE MEAN

Suppose a random sample of size ‘n’ is drawn from a given finite population of size ‘N’. Let X_{α} ($\alpha = 1, 2, \dots, N$) be the value of the variable x for the α^{th} member of the population. Then the population mean of x is $\mu = (1/ N) \sum_{\alpha} X_{\alpha}$, and the population variance is $\sigma^2 = (1/N) \sum_{\alpha} (X_{\alpha} - \mu)^2$.

Again, let us denote by x_i ($i=1, 2, \dots, n$) the value of x for the i^{th} member (i.e., the member selected at the i^{th} drawing) of the sample. The sample mean of x is then $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$. For deriving the expectation and standard error of $\bar{x}$, we consider two distinct cases:

Case I: Random Sampling With Replacement

Let us recall the following two theorems of the probability theory; (i) If $y = bx$, then $E(y) = bE(x)$, and (ii) if x and y are two random variables and z a third random variable such that $z = x + y$, then $E (z) = E(x) + E(y)$.

Based on the above information, we can write the following:

$$E(\bar{x}) = \frac{1}{n} \sum_{i=1}^n E(x_i) \text{ and}$$

$$\begin{aligned} \text{var}(\bar{x}) &= E\{\bar{x} - E(\bar{x})\}^2 \\ &= (1/n^2) E\left[\sum_i \{x_i - E(x_i)\}\right]^2 \\ &= (1/n^2) \sum_i E\{x_i - E(x_i)\}^2 + (1/n^2) \sum_{\substack{i,j \\ i \neq j}} E\{x_i - E(x_i)\} \{x_j - E(x_j)\} \\ &= (1/n^2) \sum_i \text{var}(x_i) + (1/n^2) \sum_{\substack{i,j \\ i \neq j}} \text{cov}(x_i, x_j) \end{aligned}$$

To obtain $E(x_i)$ and $\text{var}(x_i)$, we note that x_i can assume the values $X_1, X_2, \dots, X_n$, each with probability $(1/N)$.

$$\text{Hence } E(x_i) = \sum_{\alpha} X_{\alpha} P[x_i = X_{\alpha}] = \sum_{\alpha} X_{\alpha} \times (1/N) = \mu \text{ and}$$

$$\text{var}(x_i) = E(x_i - \mu)^2 = \sum_{\alpha} (X_{\alpha} - \mu)^2 \times (1/N) = \sigma^2 \text{ for each } i$$

$$\begin{aligned} \text{Again, } \text{cov}(x_i, x_j) &= E(x_i - \mu)(x_j - \mu) \\ &= \sum_{\alpha, \alpha'} (X_{\alpha} - \mu)(X_{\alpha'} - \mu) P[x_i = X_{\alpha}, x_j = X_{\alpha'}] \end{aligned}$$

Since in sampling with replacements the composition of the population remains the same throughout the sampling process, x_j can take any one of the values $X_1, X_2, \dots, X_N$, with probability $(1/N)$, irrespective of the value taken by x_i . In other words, for $i \neq j$, x_i and x_j are independent, we have:

$$P[x_i = X_{\alpha}, x_j = X_{\alpha'}] = P[x_i = X_{\alpha}] P[x_j = X_{\alpha'}] = (1/N^2)$$

$$\text{Hence, } \text{cov}(x_i, x_j) = (1/N^2) \sum_{\alpha, \alpha'} (X_{\alpha} - \mu)(X_{\alpha'} - \mu)$$

$$= (1/N^2) \sum_{\alpha} (X_{\alpha} - \mu) \sum_{\alpha'} (X_{\alpha'} - \mu) = 0$$

[Since, for each i, j ($i \neq j$), $\sum_{\alpha} (X_{\alpha} - \mu) = \sum_{\alpha'} (X_{\alpha'} - \mu)$, (being the sum of the deviations of $X_1, X_2, \dots, X_n$ from their mean is zero)].

$$\text{Hence, we have, } E(\bar{x}) = (1/n) \times n \mu = \mu$$

$$\text{and } \text{var}(\bar{x}) = (1/n^2) n \sigma^2 + (1/n^2) n(n-1) \times 0 = \sigma^2/n$$

The standard error of $\bar{x}$ is, therefore, $\sigma_{\bar{x}} = \sigma/\sqrt{n}$

Case II: Random sampling without replacement

As before for each i , $E(x_i) = \mu$ and $\text{var}(x_i) = \sigma^2$,

since here too x_i can take any one of the values $X_1, X_2, \dots, X_N$, with the same probability $(1/N)$. The covariance term, however, needs special attention.

Here, for $i \neq j$

$P[x_i = X_\alpha, x_j = X_{\alpha'}] = P[x_i = X_\alpha] P[x_j = X_{\alpha'} | x_i = X_\alpha] = (1/N)(1/(N-1))$ if $\alpha \neq \alpha'$
[since x_j can take any value except X_α , the value which is known to have been already assumed by x_i , with equal probability $1/(N-1)$] = 0 if $\alpha = \alpha'$

Hence, $\text{cov}(x_i, x_j) = (1/N(N-1)) \sum_{\substack{\alpha, \alpha' \\ \alpha \neq \alpha'}} (X_\alpha - \mu)(X_{\alpha'} - \mu) = (1/N(N-1))$

$$\sum_{\alpha} (X_\alpha - \mu) \left\{ \sum_{\alpha'} (X_{\alpha'} - \mu) - (X_\alpha - \mu) \right\} = (1/N(N-1)) \left\{ \sum_{\alpha} (X_\alpha - \mu) \sum_{\alpha'} (X_{\alpha'} - \mu) - \sum_{\alpha} (X_\alpha - \mu)^2 \right\}$$

$$= - (1/N(N-1)) N \sigma^2 = - \sigma^2 / (N-1)$$

Thus, in this case we have $E(\bar{x}) = (1/n) n \mu = \mu$

and $\text{var}(\bar{x}) = (1/n^2) \times \sigma^2 + (1/n^2) \times n(n-1) \times (-\sigma^2 / (N-1)) = (\sigma^2 / N) \{1 - (n-1)/(N-1)\}$

$$= (\sigma^2 / n) \{(N-n) / (N-1)\}$$

Hence the standard error of $\bar{x}$ is $\sigma_{\bar{x}} = (\sigma / \sqrt{n}) \sqrt{1 - \frac{N-n}{N-1}}$

In both the cases, the standard error decreases with increasing n . The standard error of the mean in sampling without replacement is, however, smaller than that in sampling with replacements. But the difference become negligible if N is very large compared to n . Also, in sampling without replacements, the standard error of the sample mean vanishes if $n = N$, which is to be expected because the sample mean now becomes a constant, i.e., the same as the population mean. However, this is not the case with sampling with replacements.

28.10 EXPECTATION AND STANDARD ERROR OF SAMPLE PROPORTION

Suppose in a population of size N , there are Np members with a particular characteristic A and Nq members with the characteristic not- A . Then p is the proportion of members in the population having the characteristic A . Let a sample of size n be drawn from the population, and let f be the number of members in the sample having characteristic A . To find the expectation and standard error of the sample proportion f/n , we adopt the following procedure.

We assign to the α^{th} member of the population the value X_α , which is equal to 1 if this member possesses the characteristic A and equal to 0 otherwise. Similarly, to the i^{th} member of the sample we assign the value x_i , which is equal to 1 if this member possesses A and is equal to 0 otherwise.

In this way, we get a variable x , which has population mean $(1/N) \sum_{\alpha} X_\alpha = p$